AEMS integration guide

The AEMS (Advanced Equipment Maintenance System) team is making APIs available specifically for external depot and vendor systems to manage and monitor orders within the AEMS system. Access to these endpoints will be restricted to systems and applications approved and onboarded as external order providers.

Document control

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2024-11-25	2.0	Self-onboarding update	OSI022
2025-03-25	3.0	Invitation code and FAQs update	SGH135

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Introduction

This guide provides the necessary information for external vendors to integrate with AEMS APIs from a functional perspective. The detailed structures, syntax, and data contracts for each API are available as downloads on our product page.

Product Page: https://developer.maersk.com/api-catalogue/aems-maintenance-repair

Technical details are not included in this document, so it is essential to reference above page for comprehensive usage guidelines.

Self-onboard

We have integrated AEMS APIs into the Maersk Developer Portal which enables external and internal developers requesting access and interacting with Maersk APIs.

We encourage existing consumers who received credentials over mail, to follow the Getting Started guide to get a new set of credentials. These new credentials can be used for both stage and production environment. See the **Getting Started** and **Get authentication token** sections in this document for more details.

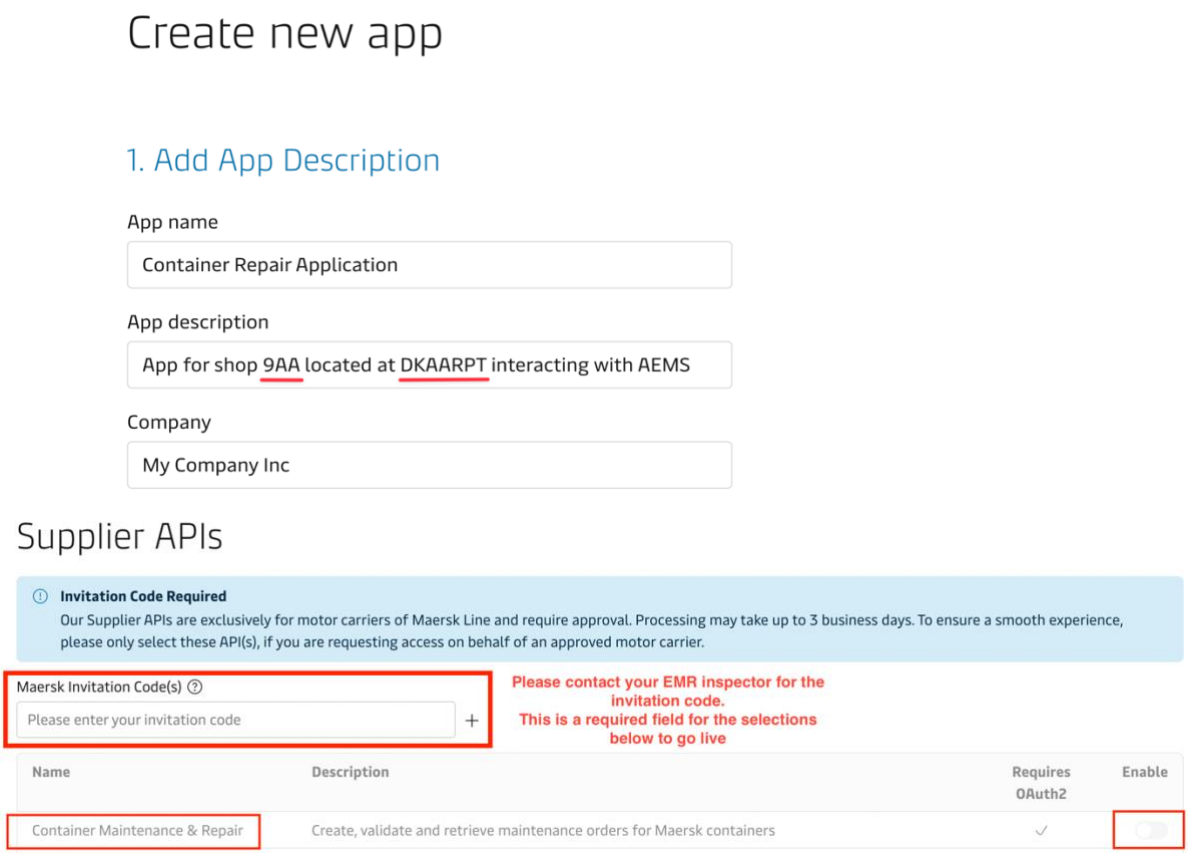
Getting Started

To get stated using AEMS APIs you would need to visit and read **Getting Started** guide. Please read this section guide before you begin your registration. Not following below guide may result in your application not getting approved.

Register a user in Maersk Developer Portal
Once registering a user please make sure to use the authorized email associated to shop (if possible). This helps the AEMS API approvers authenticating the right consumers.

Create a new application
When you "Create a new application', you are registering the consumer which will be used to interact with the AEMS. A good name could be the name of the application sending estimates.

Please include shop code and location code as a part of the description as visualized below.



FAQ

Q. How to obtain the shop code, location code or the invitation code?
A. Please ***contact your EMR inspector*** for obtaining the shop code, location code or the invitation code

Once the access request has been created and approved, you can use the application credentials to call Maersk Authentication API, to obtain an *access token*. The *access token* will be used as authentication to call any AEMS subsequent API.

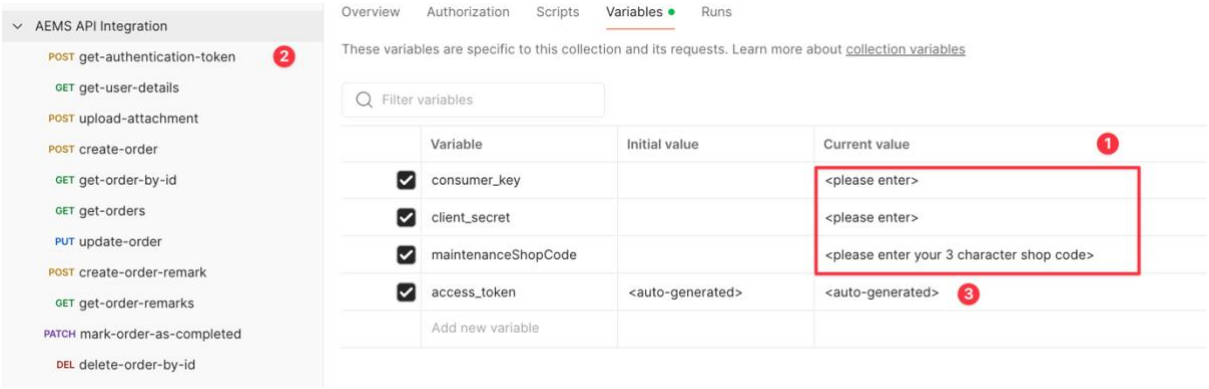
Example Credentials

Consumer-Key	`N0NRu0JBU40BRCLSGqxso4G67Oj0omIm`
Client Secret	`yssTIO1Zh9SMk5Ab`

Get authentication token

In the sample Postman script, we have prepared a function for obtaining an authentication token for you. [Download postman script]

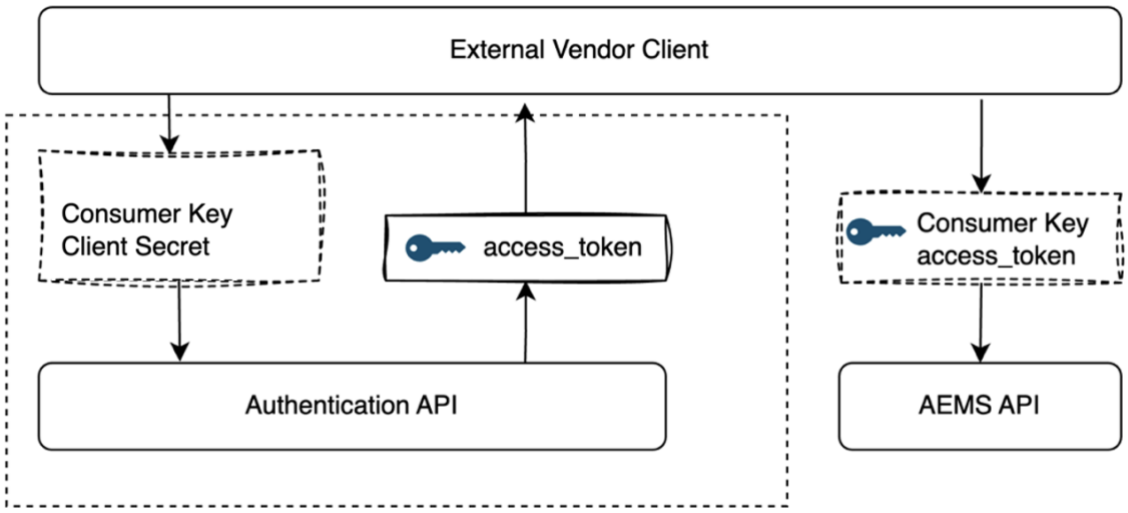
First, please import the script into Postman and replace the collection variables with your credentials, as visualized in **(1)**. Make sure to save file as your credentials are not updated!



The mainteananceShopCode will be the 3-character code identifying your shop.

Next, please navigate to *get-authentication-token* **(2)** and hit "*Send"* in the upper right corner. This will send a request to the Authentication API using the entered credentials to obtain an *access token*.

The access token received in the response is automatically copied into the access_token variable by the postman script.

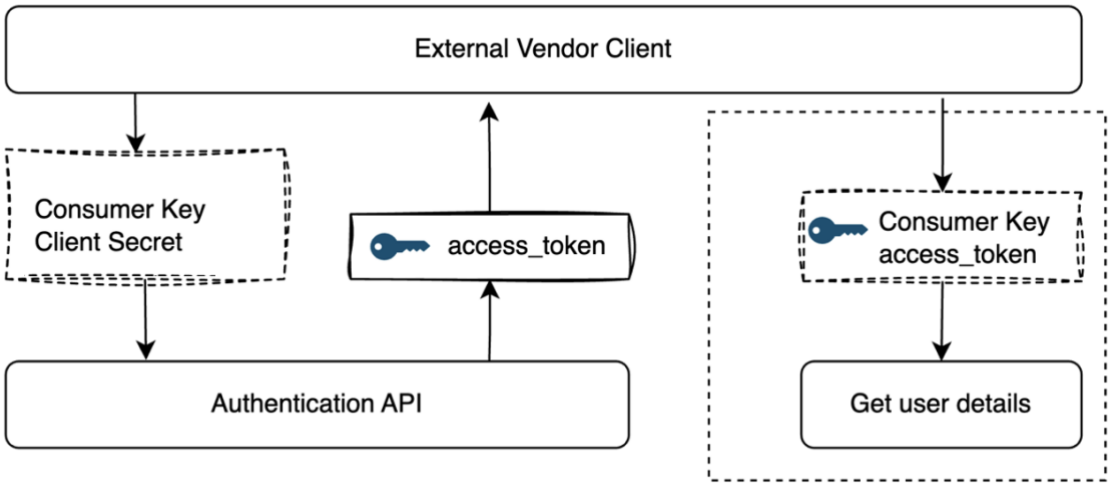


Now, please navigate to any other AEMS call in the left pane and hit "Send". These functions all uses the entered *consumer_key* variable and the *access_token*, to authenticate the calls.

Testing successful integration

Once your client has successfully authenticated, you can verify the setup by testing the **user-details** endpoint. This will ensure that the client is authorized and has access to the necessary shop resources.

Sample script: `get-user-details`



The **get-user-details** Postman script can be used to test the authentication flow and validate that the client configuration is correct, or you can use the following cURL.

Client Details Request:
To initiate this test, make a request to the **user-details** endpoint. This request will check if the client is properly authenticated and authorized to interact with the shop resources.

Client Details Response:
Upon a successful request, you will receive a response that includes information about the client and the shop resources they have access to. The response should include the agreed-upon shop code(s) that were defined during the onboarding phase.


```
{
  "username": "xyz-client",
  "emailAddress": "kim.jensen@myvendor.com",
  "companyName": "MyVendor A/S",
  "isExternal": true,
  "roles": [
    "Shop"
  ],
  "maintenenceShopCodes": [
   {
     "maintenenceShopCode": "xyz",
     "maintenenceShopName": "MyVendor Repairs"
   }]
}
```


Make sure that:

- Ensure that the returned **shop code(s)** match the values provided during the onboarding process.
- Contact email matches your company email or an email of an employee or unit that takes care of maintenance orders.
- IsExternal property is set to `true`.

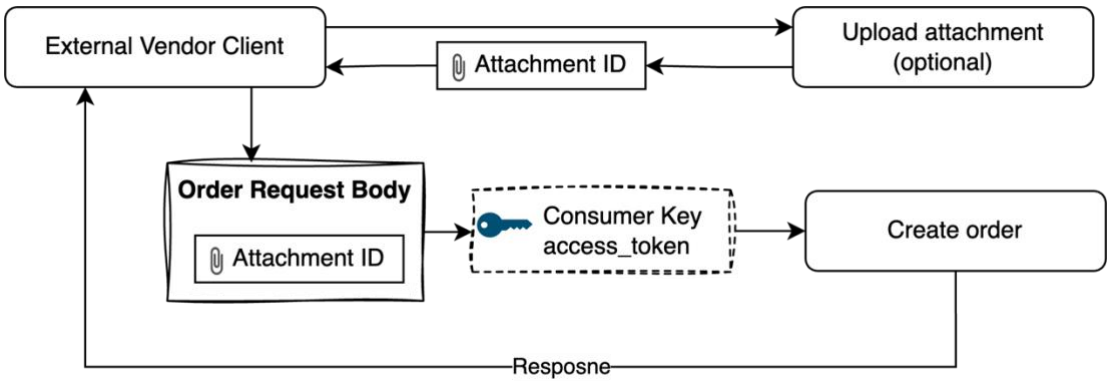
Order management

This section will take you through the flow of order management from order creation to order completion. Additional endpoints not covered in this section can be found in the OpenAPI Specification (OAS).

Creating an Order

This section explains how to create an order with all the required details and optionally attach an uploaded resource. As with all API operations, an **access token** is required to call both the order and attachment endpoints.

Sample script: `create-order`



Order attachments (optional)
In AEMS, uploading a resource (such as images or documents) is handled **separately** from the order creation process. After successfully uploading a resource, you will receive a unique ID of the uploaded resource (with file extension), which can then be used to link the resource to the order.

Attachment ID: 1cd3cf29-d588-4cb1-9dd3-6f17a506c666.png

To include an attachment in the order, add the Attachment ID to the appropriate arrays in the order request body, as shown below. As specified In the OpenAPI Specification, attachments can be put on either **maintenance order lines** or to the **order itself**, depending on your needs.

```
"attachedImages" [
  {
    "contentType": "image/jpeg",
    "extension": ".JPG",
    "identifier": "1cd3cf29-d588-4cb1-9dd3-6f17a506c666.JPG",
    "name": "OrderAttachment.JPG",
    "path": "1cd3cf29-d588-4cb1-9dd3-6f17a506c666.JPG"
  }
```

Order creation
Once the order request body has been properly filled out according to the **OAS specification**, you can submit the request to create the order.

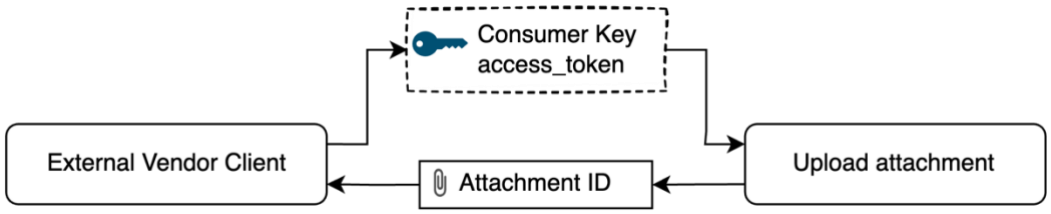
Make sure that:

- All required fields are populated.
- The Attachment ID(s) are correctly inserted if resources are being attached.

Upload an Order Attachment

Attachments are uploaded through a **resource-based API** that operates separately from the order creation process. Clients can upload files such as images, PDFs, or other lightweight content, and upon successful upload, a **GUID** (unique file ID) will be returned.

Sample Script: `upload-attachment`



Once the file has been uploaded and stored, the system will respond with a unique **Attachment ID** that can be used to reference the file in subsequent API requests, such as when creating or updating an order.

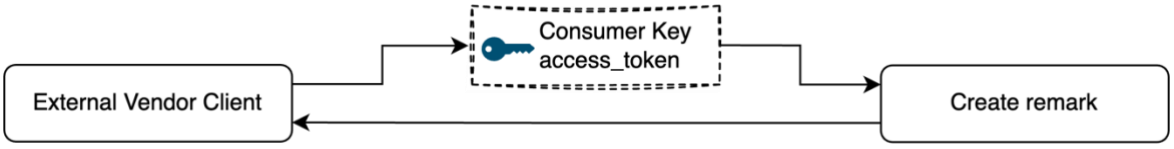
For example:

- **Attachment ID**: 1cd3cf29-d588-4cb1-9dd3-6f17a506c666.png

Creating Order Remarks

Order remarks can either be added during the **order creation process** or separately via a dedicated endpoint, as demonstrated in the sample script.

Sample Script: `create-order-remark`



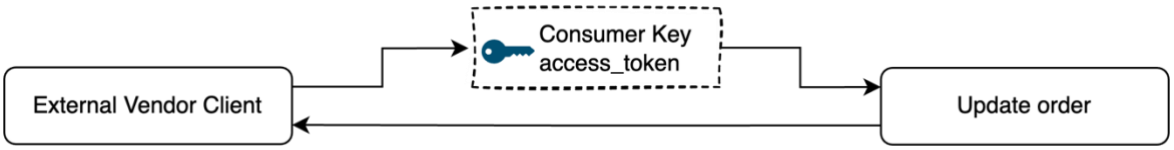
Adding Remarks During Order Creation
If remarks need to be added while creating an order, simply include them in the **maintenanceOrderRemarks** section within the JSON request body, as outlined in the **OpenAPI Specification (OAS)**.

Adding Remarks After Order Creation
To add remarks separately after the order has been created, you can use the **create-order-remark** endpoint by providing the relevant **Order ID**.

Updating Existing Order Details

If an existing order needs to be modified—whether due to non-approval, missing attachment documentation, or incorrect information—you can use the **update-order** API. Just like with order creation, a validation process will be triggered when submitting the update request to ensure all data is correct.

Sample script: `update-order`

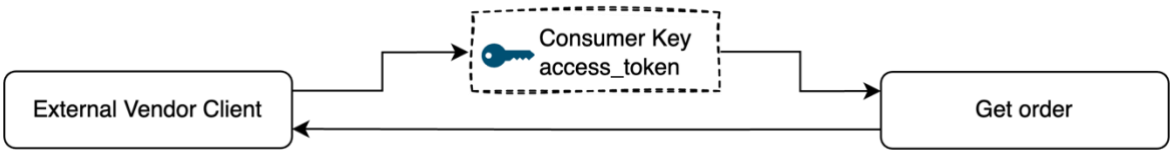


Make sure to review and validate the order details before submitting the update to prevent further issues or rejections.

Retrieving Order Details

Tracking an order's status is essential to determine when it has been approved by Maersk GSC staff or to check any other order state. The **GET order details** endpoint provides crucial information, including the status, pricing, and other vital details about an existing order.

Sample Script: `get-order-by-id`

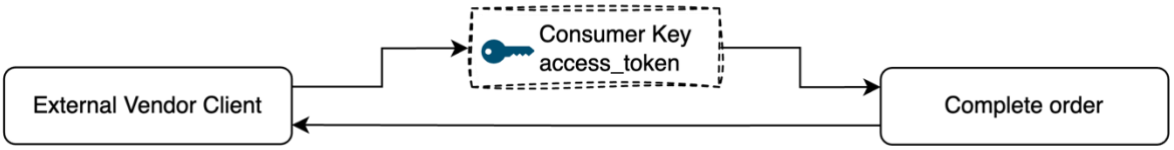


By calling this endpoint, you can retrieve the full order details, allowing you to monitor the progress and state of the order in real-time.

Marking an Order as Completed

Once an order has been approved and the repairs or tasks have been completed by the vendor, the order can be marked as completed using the **mark as complete** endpoint.

Sample script: `mark-order-as-completed`



This action signifies that all work related to the order has been finished and no further updates are required.

Request URIs:

Maintaining and monitoring orders	https://api-stage.maersk.com/aems/orders/
Up- and downloading order attachments	https://api-stage.maersk.com/aems/blob-manager/
User/Client identity	https://api-stage.maersk.com/aems/identity/

Staging https://api-stage.maersk.com/aems/
Production https://api.maersk.com/aems/

Access from stage to production environment will be elevated once the integration test has been performed.

FAQ

I do not have documentation on fields for orders.

Please make sure you have downloaded and parsed the OAS specifications, found at our product page in Maersk Developer Portal. These contains all data types and descriptions.

Client authentication failed

Once authenticating you are getting **error: invalid_client, Client authentication failed**. This can be due an rare seen bug in the process of creating a consumer key in the system.

To fix this we would like you to try signing in to Maersk Developer Portal, and navigating to your Application Entry. From here you should be able to add a new consumer key by clicking **"Add Consumer Key".**

This will send a new request to the AEMS approval team. Once this has been approved, kindly try reuse the new consumer key and new secret.

User does not have access to shop

You are doing all good. Once your request has been approved, the AEMS team needs to map your consumer key to your respective shop, to ensure every consumer has the correct permissions.

If you see the **Status code 403 forbidden** - User 'xyz-client' does not have access to the shop XYZ, please reach out to the AEMS developer team.

How do I go from stage to production

Once you have successfully integrated and tested with our staging environment, you can access production URLs, which you will find in the **Request URIs** section of this document.

If you requested credentials via Maersk Developer Portal following **Getting Started** section, your credentials will work for both stage and production. Otherwise, your Client ID formatted like 'xyz-client', you would have to follow the **Getting Started** section to get a new pair of credentials.

Note, we require at least 50 successfully created orders in staging environment before we enable your shop creating orders in production.